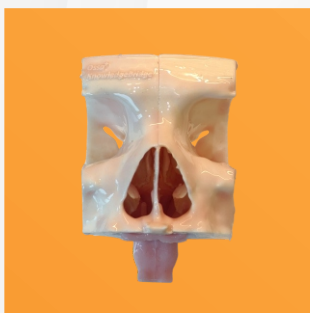


High-Fidelity 3D Surgical Simulation Workshops in Otolaryngology

2026



Dear Colleagues,

We are pleased to present the High-Fidelity 3D Surgical Simulation Workshops in Otolaryngology – 2026, a comprehensive series of hands-on training programs developed using anatomically accurate 3D simulation models.

The workshop portfolio covers key areas of otology, neurotology, rhinology, and laryngology, including Basic Temporal Bone Dissection, Facial Nerve Surgery, Tympanoplasty, Cochlear Implant Surgery, Advanced Temporal Bone Dissection, Paranasal Sinus Surgery, Balloon Sinuplasty and Eustachian Tube Dilatation, and Laryngeal Surgery.

Each workshop is conducted as a one-day, faculty-led, hands-on program, using high-fidelity simulation models that closely replicate external and internal anatomy. The training format includes live demonstration of each procedural step by faculty, followed by supervised participant practice at fully equipped workstations designed to simulate real laboratory and operating room conditions. Interactive discussions and question-answer sessions are incorporated where described, to reinforce anatomical understanding and procedural concepts.

These workshops are intended to provide participants with structured, reproducible training in a controlled environment, supporting skill development and procedural confidence before undertaking live surgery. We invite you to review the enclosed brochure for detailed information on each workshop and enrol in upcoming workshop in 2026.

Warm Regards,

Dr. Prashant Naik

MS(ENT), DLO

Co-Founder

Ossa Plus Simulation LLP

Surgical Simulation

Shashikumar Sambhoo

Co-Founder

Ossa Plus Simulation LLP

KnowledgeBridge International Pvt. Ltd.

Medical Education & Surgical Simulation

Basic 3D Temporal Bone Dissection workshop

Clinical Context & Rationale

Why this workshop?

- Temporal bone anatomy is highly complex and compact
- Safe otologic surgery requires precise 3-D anatomical orientation
- Cadaveric dissection before live surgery improves safety and confidence
- Simulation-based training provides reproducible learning



Procedures Covered

- Identification of external anatomical landmarks
- Cortical mastoidectomy
- Posterior tympanotomy
- Cochleostomy
- Facial nerve decompression
- Cochlear implant dummy electrode insertion

Anatomical /

Simulation Model Overview

- Bone model with accurate external and internal anatomical details including
- Mastoid air cells
- Middle ear cleft
- Facial nerve canal
- Cochlea & semicircular canals
- Sigmoid sinus, and internal acoustic canal



- Atticotomy
- Modified radical mastoidectomy
- Labyrinthectomy
- Translabyrinthine approach
- Endolymphatic sac approach
- Bill's island technique



Program Duration & Format

One-day hands-on workshop

Workstation & Infrastructure

- Operating microscope
- High-speed microdrill with cutting & diamond burs
- Suction & irrigation system
- Temporal bone holders
- Complete dissection instrument set

Training Methodology

- Step-by-step live faculty demonstration
- Real-time dissection on 3D temporal bone models
- One-to-one mentoring during practice
- Continuous faculty feedback



Key Takeaways

Imparts foundational dissection skills and confidence in handling temporal bone anatomy



Awards



Recognition



Participation



Acknowledgement



Skills You Will Learn

- ▶ Identification of external anatomical landmarks
- ▶ Systematic temporal bone dissection skills

Facial Nerve Workshop

Clinical Context & Rationale

Why this workshop?

- The facial nerve follows a complex and intricate path through the temporal bone
- Facial nerve surgery therefore demands exceptional precision and control
- Developing these skills requires meticulous and systematic dissection training
- Anatomically accurate cadaveric bone replicas provide the ideal learning platform

Anatomical / Simulation Model Overview

- Artificial temporal bone model with exact internal and external anatomy
- Complete intricate pathway of the facial nerve



Procedures Covered

- Decompression from first genu to stylomastoid foramen
- Re-routing of the facial nerve
- End-to-end nerve anastomosis
- Various types of nerve grafting



Program Duration & Format

One-day hands-on workshop

Workstation & Infrastructure

- Temporal bone dissection instruments,
- Diamond burrs of different sizes,
- Nerve graft pieces,
- Nerve suturing materials

Training Methodology

- Step-by-step live faculty demonstration
- Real-time dissection on individual models under expert guidance
- One-to-one mentoring during practice
- Continuous faculty feedback



Key Takeaways

Imparts skills and confidence for facial nerve surgeries



Awards



Recognition



Participation



Acknowledgement



Skills You Will Learn

- ▶ Blue-lining the facial nerve
- ▶ Facial nerve decompression
- ▶ Facial nerve re-routing
- ▶ End-to-end anastomosis
- ▶ Nerve grafting techniques

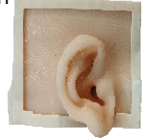
Tympanoplasty Workshop

Clinical Context & Rationale Why this workshop?

- Tympanoplasty demands high surgical precision to avoid complications and failure
- Skill development requires structured, hands-on surgical practice
- Practical training significantly improves outcomes and success rates

Anatomical / Simulation Model Overview

- Artificial temporal bone model with tympanic membrane
- Having moderate central perforation
- Some models with absent or eroded incus



Procedures Covered

- Endomeatal incision
- Inlay graft technique
- Repositioning of flap
- Ossiculoplasty when incus is absent



Program Duration & Format One-day hands-on workshop

Workstation & Infrastructure

- Temporal bone dissection instruments
- Microear instruments
- artificial grafts, gelfoam
- myringotomy knife, grommets
- separate incus for sculpting

Training Methodology

- Step-by-step live faculty demonstration
- Real-time dissection on individual models under expert guidance
- One-to-one mentoring during practice
- Continuous faculty feedback



Key Takeaways

Imparts skills and confidence for tympanoplasty surgery



Awards



Recognition



Participation



Acknowledgement



Skills You Will Learn

- ▶ Myringotomy
- ▶ Grommet insertion
- ▶ Freshening perforation edges
- ▶ Tympanomeatal flap elevation
- ▶ Graft placement
- ▶ Ossiculoplasty

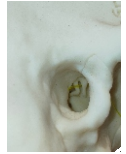
Cochlear Implant Surgery Workshop

Clinical Context & Rationale Why this workshop?

- Cochlear implant surgery requires thorough anatomical understanding,
- Precise surgical technique,
- Atraumatic electrode insertion

Anatomical / Simulation Model Overview

- Artificial temporal bone model with mastoid air cells
- Facial recess, round window niche
- Cochlea, and facial nerve pathway



Procedures Covered

- Mastoid antrotomy
- Facial recess approach
- Round window approach
- Stimulator well creation
- Electrode array tunnelling
- Pediatric cochlear implantation techniques



Program Duration & Format One-day hands-on workshop

Workstation & Infrastructure

- Temporal bone instruments,
- Microscope, suction and irrigation systems
- Microdrill with cutting and diamond burrs
- Dummy electrodes, bone holders, trays

Training Methodology

- Step-by-step live faculty demonstration
- Real-time dissection on individual models under expert guidance.
- One-to-one mentoring during practice
- Continuous faculty feedback



Key Takeaways

Imparts technical skills and confidence for cochlear implantation



Awards



Recognition



Participation



Acknowledgement



Skills You Will Learn

- ▶ Temporal bone anatomy orientation
- ▶ Identification of surgical landmarks
- ▶ Facial recess dissection
- ▶ Round window electrode insertion

Advanced Temporal Bone Dissection Workshop

Clinical Context & Rationale

Why this workshop?

- Mastery of basic mastoid surgery is the foundation for advanced procedures
- Temporal bone surgery requires progressively structured dissection training
- Advanced skills are built through guided, hands-on practice

Anatomical /

Simulation Model Overview

- Artificial temporal bone model with real-size inner ear
- Complete facial nerve, internal acoustic canal
- Sigmoid sinus, and dura



Procedures Covered

- Facial nerve decompression
- Facial recess approach
- Labyrinthectomy
- Translabyrinthine approach
- Bill's island technique
- Endolymphatic sac approach



Program Duration & Format

One-day hands-on workshop

Workstation & Infrastructure

- Basic temporal bone dissection
- Microear instruments

Training Methodology

- Step-by-step live faculty demonstration
- Real-time dissection on individual models under expert guidance.
- One-to-one mentoring during practice
- Continuous faculty feedback



Key Takeaways

Enhances skills and confidence for advanced temporal bone surgeries



Awards



Recognition



Participation



Acknowledgement



Skills You Will Learn

- ▶ Advanced temporal bone surgical techniques

Paranasal Sinuses Workshop

Clinical Context & Rationale

Why this workshop?

- Paranasal sinus anatomy shows wide anatomical variation
- Endoscopic sinus surgery demands strong three-dimensional orientation
- Accurate spatial understanding is critical for safe surgical navigation

Anatomical / Simulation Model Overview

- Artificial paranasal sinus model with septum
- Turbinates, uncinate process, bulla ethmoidalis
- Sinuses with ostia, and soft tissue mucosa



Procedures Covered

- Septoplasty
- Uncinectomy
- Maxillary antrotomy
- Anterior and posterior ethmoidectomy
Frontal recess approach
- Sphenoidectomy
Transphenoidal approach



Program Duration & Format

One-day hands-on workshop

Workstation & Infrastructure

- Artificial PNS model
- Endoscopic instruments
- Endoscopes
- Monitor tower

Training Methodology

- Step-by-step live faculty demonstration
- Real-time dissection on individual models under expert guidance.
- One-to-one mentoring during practice
- Continuous faculty feedback



Key Takeaways

Imparts skills and confidence in paranasal sinus surgeries



Awards



Recognition



Participation



Acknowledgement



Skills You Will Learn

- ▶ Endoscopic sinus surgical skills

Balloon Sinuplasty & Eustachian Tube Dilatation Workshop

Clinical Context & Rationale Why this workshop?

- Balloon-based sinus and Eustachian tube procedures demand high precision
- Clear anatomical orientation is essential for safe and effective intervention
- Hands-on training is critical to achieve procedural mastery

Anatomical / Simulation Model Overview

- Artificial paranasal sinus model with ergonomically designed soft tissues for balloon catheterization



Procedures Covered

- Balloon sinuplasty of frontal sinuses
- Balloon sinuplasty of maxillary sinuses
- Eustachian tube dilatation



Program Duration & Format

Single-day hands-on program with didactic lectures and video demonstrations

Workstation & Infrastructure

- Artificial PNS model
- Balloon catheter instruments
- Nasal endoscopes
- Light source, camera, monitor tower

Training Methodology

- Step-by-step live faculty demonstration
- Real-time dissection on individual models under expert guidance.
- One-to-one mentoring during practice
- Continuous faculty feedback



Key Takeaways

**Builds technical skills and confidence for
balloon-based procedures**



Awards



Recognition



Participation



Acknowledgement



Skills You Will Learn

- ▶ Balloon sinuplasty techniques
- ▶ Eustachian tube catheterization

Larynx Workshop

Clinical Context & Rationale Why this workshop?

- Laser technology has revolutionized modern laryngeal surgery
- These procedures now demand high levels of precision and control
- Refined motor skills are essential for safe day-care surgical outcomes

Anatomical / Simulation Model Overview

- Artificial 3D larynx model with real-size anatomy
- Replaceable glottic cassettes simulating various lesions



Procedures Covered

- Vocal cord nodule excision
- Partial cordectomy
- Laryngeal web excision



Program Duration & Format One-day hands-on workshop

Workstation & Infrastructure

- Artificial larynx model
- Endoscopic instruments
- Endoscopes, monitor tower
- Laser machines

Training Methodology

- Step-by-step live faculty demonstration
- Real-time dissection on individual models under expert guidance.
- One-to-one mentoring during practice
- Continuous faculty feedback



Key Takeaways

Imparts skills and confidence for office-based laryngeal surgeries



Awards



Recognition



Participation



Acknowledgement

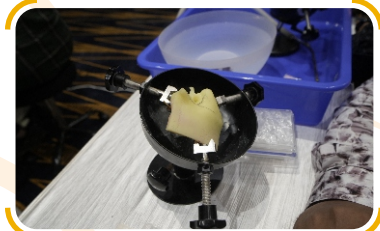


Skills You Will Learn

- ▶ Precision laryngeal surgical skills
- ▶ Laser-based cord procedures



Temporal Bone Workshop Glimpses



3D Surgical Simulation Workshops

**Basic
3D Temporal
Bone Dissection**

Facial Nerve

**Advanced
Temporal Bone
Dissection**

Tympanoplasty

**Paranasal
Sinuses**

Larynx

**Cochlear
Implant
Surgery**

**Balloon Sinuplasty &
Eustachian Tube Dilatation**